Reg. No.										

MANIPAL INSTITUTE OF TECHNOLOGY

MANIPAL

(A constituent unit of MAHE, Manipal)

SEVENTH SEMESTER BTECH. (E & C) DEGREE END SEMESTER EXAMINATION

DECEMBER 2021-JANUARY 2022

SUBJECT: Information Theory and Coding (ECE -4075)

TIME: 75 minutes **MAX. MARKS: 20**

Instructions to candidates

- Answer **ALL** questions.
- Missing data may be suitably assumed.

Q. No.	Questions	Marks
1A.	Construct a minimum variance Huffman code for the source shown in the following table using the code alphabet X= {0,1,2}. S: s1 s2 s3 s4 s5 s6 s7 s8 s9 s10 P(s_i): 0.20 0.18 0.12 0.10 0.10 0.08 0.06 0.06 0.06 0.04 Find efficiency and redundancy of this code.	4
1B.	Let S be zero memory source with source alphabet $S = \{s_i\}, i = 1,2, \dots, q$, and symbol probabilities $P_1, P_2, \dots, P_q$. Construct a new zero-memory source S' with twice as many symbols, $S' = \{s_i'\}, i = 1,2, \dots, 2q$. Let P_i', the symbol probabilities for the new source, defined by $P_i' = (1 - \epsilon)P_i, \quad i = 1,2, \dots, q$ $P_i' = \epsilon P_{i-q} \quad i = q+1, q+2, \dots, 2q.$ Write $H(S')$ in terms of $H(S)$.	3
1C.	Describe Uniform channel. Describe the mutual information of r-ary symmetric channel with the overall probability of error p. Also identify the capacity of this channel.	3
2A.	A source with 26 alphabets (A-Z) emits a message "**shannon**". Generate a binary code for this message using Adaptive Huffman coding algorithm. Show updated tree properly.	5
2B	Explain encoding procedure and update procedure of Adaptive Huffman coding.	3
2C	Justify two real time scenarios for the additivity of Mutual information.	2